



YIRAN LU

DATA SCIENTIST

DOCTOR IN NANOPHYSICS

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✉ yiranls@outlook.com

📍 Grenoble, France

in [yiran-lu-66424012a](#)

📍 Grenoble and surrounding area

SKILLS

Techniques

- Python, C++, SQL, CUDA
- ML & Deep Learning (Tensorflow)
- Statistics Analysis
- Fourier Analysis
- Git/Github, Docker, DevOps
- Data Structure & Algorithms
- Database

Soft Skills

- Teamwork & Autonomy
- Analytical skills
- Rigorous and curious
- Agile (SCRUM)

Languages

- **Chinese** : Native
- **English** : Level C1/C2
- **French** : Level B2/C1

EDUCATION

2025-2026
(6 months of intensive training)

DATA SCIENTIST

Campus Numérique in the Alpes

[\(Link to the program\)](#)

2021 - 2025

PHD IN NANOPHYSICS

Université Grenoble-Alpes

2019 - 2021

MASTER OF NANOSCALE ENGINEERING

Université Claude-Bernard Lyon 1

2015 - 2019

BACHELOR OF ENGINEERING

University of Science et Technologie Beijing

Analytical Doctor (PhD) leveraging 4 years of rigorous data processing and **MLOps** skills. Expertise in **Optimization, Fourier Analysis, and HPC (Python/CUDA)** for production-ready solutions.

PROFESSIONAL EXPERIENCE

Institut NEEL, Grenoble

* Nov. 2021 to Apr. 2025

PhD Candidate in Physics

- Engineered and deployed high-performance computing (HPC) solutions using Python and CUDA to process and analyze GB-scale datasets from advanced imaging techniques.
- Developed and implemented sophisticated Optimization Algorithms and custom Mathematical Methods (including Fourier-based reconstruction techniques) for materials property quantification, contributing code to an established open-source software package.
- Established robust data governance and management strategies to organize and process experimental data, ensuring a reproducible environment and accelerating analysis cycles.

CEAGrenoble

* Mar. to Aug. 2021

Research Intern

- Optimized and developed high-throughput data processing pipelines using Python (including Numba for performance), successfully accelerating calculation scripts from minutes to mere seconds.
- Resolved critical mathematical errors and bugs within legacy analysis software, ensuring the reliability of results, while designing Exploratory Data Analysis (EDA) tools for enhanced data interpretation.

Institut Nanotechnologie de Lyon

* Jan. to Mar. 2020

Research Intern

- Engineered and executed the physical setup of a complex optical path using a spatial light modulator for precise wave-front shaping.
- Utilized MATLAB simulation software for complex algorithm implementation (e.g., Fourier-based optimization methods) and data analysis to inspect experimental results and determine optimal system parameters.

HOBBIES

Competitive Programming (C++) / **Chess** / Programming
(Common Lisp) / Guitar / Hiking / Via-ferrata